The Invariance Principle — problem 1.28

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n numbers are written on a blackboard. In one step you may erase any two of the numbers, say a and b, and write, instead $(a+b)/4$. Repeating this step $n-1$ times, there is one number left. Prove that, initially, if there were n ones on the board, at the end, a number, which is not less than $1/n$ will remain.